

ATTACHMENT A

Revise the Beaver Valley Unit No. 1 Technical Specifications as follows:

Remove Page

2-6
2-7
2-7a
2-10
3/4 3-22
3/4 3-22a
3/4 3-23
3/4 3-24
3/4 3-24a

Insert Page

2-6
2-7
2-7a
2-10
3/4 3-22
3/4 3-22a
3/4 3-23
3/4 3-24
3/4 3-24a

TABLE 2.2-1

REACTOR TRIP SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
1. Manual Reactor Trip	Not Applicable	Not Applicable
2. Power Range, Neutron Flux	Low Setpoint - $\leq 25\%$ of RATED THERMAL POWER High Setpoint - $\leq 109\%$ of RATED THERMAL POWER	Low Setpoint - $\leq 27.3\%$ RATED THERMAL POWER High Setpoint - $\leq 111.3\%$ of RATED THERMAL POWER
3. Power Range, Neutron Flux, High Positive Rate	$\leq 5\%$ of RATED THERMAL POWER with a time constant ≥ 2 seconds	$\leq 6.3\%$ of RATED THERMAL POWER with a time constant ≥ 2 seconds
4. Power Range, Neutron Flux, High Negative Rate	$\leq 5\%$ of RATED THERMAL POWER with a time constant ≥ 2 seconds	$\leq 6.3\%$ of RATED THERMAL POWER with a time constant ≥ 2 seconds
5. Intermediate Range, Neutron Flux	$\leq 25\%$ of RATED THERMAL POWER	$\leq 31.1\%$ of RATED THERMAL POWER
6. Source Range, Neutron Flux	$\leq 10^5$ counts per second	$\leq 1.4 \times 10^5$ counts per second
7. Overtemperature ΔT	See Note 1	See Note 3
8. Overpower ΔT	See Note 2	See Note 3
9. Pressurizer Pressure--Low	≥ 1945 psig	≥ 1934 psig
10. Pressurizer Pressure--High	≤ 2385 psig	≤ 2394 psig
11. Pressurizer Water Level--High	$\leq 92\%$ of instrument span	$\leq 93.9\%$ of instrument span
12. Loss of Flow	$\geq 90\%$ of design flow* per loop	$\geq 89.2\%$ of design flow* per loop

*Design flow is 88,500 gpm per loop.

TABLE 2.2-1 (Continued)

REACTOR TRIP SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
13. Steam Generator Water Level-Low-Low	$\geq 12\%$ of narrow range instrument span-each steam generator	$\geq 10.7\%$ of narrow range instrument span-each steam generator
14. Steam/Feedwater Flow Mismatch and Low Steam Generator Water Level	$\leq 40\%$ of full steam flow at RATED THERMAL POWER coincident with steam generator water level $\geq 25\%$ of narrow range instrument span-each steam generator	≤ 43.4 of full steam flow at RATED THERMAL POWER coincident with steam generator water level ≥ 23.1 of narrow range instrument span-each steam generator
15. Undervoltage-Reactor Coolant Pumps	≥ 2750 volts-each bus	≥ 2687 volts-each bus
16. Underfrequency-Reactor Coolant Pumps	≥ 57.5 Hz - each bus	≥ 57.4 Hz - each bus
17. Turbine Trip A. Auto stop oil pressure	45 psig	± 5 psig
B. Turbine Stop Valve	$\geq 1\%$ open	$\geq 1\%$ open
18. Safety Injection Input from ESF	Not Applicable	Not Applicable
19. Reactor Coolant Pump Breaker Position Trip	Not Applicable	Not Applicable
20. Reactor Trip System Interlocks		
A. Intermediate Range Neutron Flux, P-6	$\geq 1 \times 10^{-10}$ Amps	$\geq 6 \times 10^{-11}$ Amps

TABLE 2.2-1 (Continued)

REACTOR TRIP SYSTEM INSTRUMENTATION TRIP SETPOINTS

20. (Continued)

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
B. Power Range Neutron Flux, P-8	$\leq 30\%$ RATED THERMAL POWER	$\leq 32.3\%$ RATED THERMAL POWER
C. Power Range Neutron Flux, P-9	$\leq 49\%$ RATED THERMAL POWER	$\leq 51.3\%$ RATED THERMAL POWER
D. Power Range Neutron Flux, P-10 (Input to P-7)	10% RATED THERMAL POWER	$\geq 7.7\%$ and $\leq 12.3\%$ RATED THERMAL POWER
E. Turbine Impulse Chamber Pressure, P-13 (Input to P-7)	$\leq 10\%$ of RTP Turbine Impulse Pressure Equivalent	$\leq 12.3\%$ of RTP Turbine Impulse Pressure Equivalent

TABLE 2.2-1 (CONTINUED)

REACTOR TRIP SYSTEM INSTRUMENTATION TRIP SETPOINTS

NOTATION

NOTE 2: Overpower $\Delta T \left(\frac{1}{1+\tau_4 S} \right) \leq \Delta T_0 \left[K_4 - K_5 \left(\frac{\tau_3 S}{1+\tau_3 S} \right) \left(\frac{1}{1+\tau_5 S} \right) T - K_6 \left[T \left(\frac{1}{1+\tau_5 S} \right) - T'' \right] - f(\Delta I) \right]$

where: ΔT_0 = Indicated ΔT at RATED THERMAL POWER

T = Average temperature, °F

T'' = Indicated T_{avg} at RATED THERMAL POWER $\leq 576.3^\circ\text{F}$

K_4 = 1.07

K_5 = 0.02/°F for increasing average temperature

K_6 = 0.00128 for $T > T''$; $K_6 = (0)$ for $T \leq T''$

$\frac{\tau_3 S}{1+\tau_3 S}$ = The function generated by the rate lag controller for T_{avg} dynamic compensation.

τ_3 = Time constant utilized in the rate lag controller for T_{avg} , $\tau_3 = 10$ secs.

$\frac{1}{1+\tau_4 S}$ = Lag compensator on measured ΔT

τ_4 = Time constant utilized in the lag compensator for ΔT , ≤ 2 secs.

$\frac{1}{1+\tau_5 S}$ = Lag compensator on measured T_{avg}

τ_5 = Time constant utilized in the lag compensator for $T_{avg} \leq 2$ secs.

S = Laplace transform operator.

$f(\Delta I)$ = 0 for all ΔI

NOTE 3: The channel's maximum trip point shall not exceed its computed trip point by more than 3.4 percent.

TABLE 3.3-4

ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
1. SAFETY INJECTION, TURBINE TRIP AND FEEDWATER ISOLATION		
a. Manual Initiation	Not Applicable	Not Applicable
b. Automatic Actuation Logic	Not Applicable	Not Applicable
c. Containment Pressure--High	≤ 1.5 psig	≤ 2.4 psig
d. Pressurizer Pressure-Low	≥ 1845 psig	≥ 1830 psig
e. Steamline Pressure-Low	≥ 500 psig steam line pressure	≥ 488 psig steam line pressure

TABLE 3.3-4 (Continued)

ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

FUNCTIONAL UNIT	TRIP SETPOINT	ALLOWABLE VALUES
1.1 SAFETY INJECTION-TRANSFER FROM INJECTION TO THE RECIRCULATION MODE		
a. Manual Initiation	Not Applicable	Not Applicable
b. Automatic Actuation Logic Coincident with Safety Injection Signal	Not Applicable	Not Applicable
c. Refueling Water Storage Tank Level-Low	18'8-1/2"	$\geq 18'2-1/2"$ and $\leq 19'2-1/2"$
d. Refueling Water Storage Tank Level - Auto QS Flow Reduction	9'6"	$\geq 9'0"$ and $\leq 10'0"$

TABLE 3.3-4 (Continued)

ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
2. CONTAINMENT SPRAY		
a. Manual Initiation	Not Applicable	Not Applicable
b. Automatic Actuation Logic	Not Applicable	Not Applicable
c. Containment Pressure--High-High	≤ 8.0 psig	≤ 8.9 psig
3. CONTAINMENT ISOLATION		
a. Phase "A" Isolation		
1. Manual	Not Applicable	Not Applicable
2. From Safety Injection Automatic Actuation Logic	Not Applicable	Not Applicable
b. Phase "B" Isolation		
1. Manual	Not Applicable	Not Applicable
2. Automatic Actuation Logic	Not Applicable	Not Applicable
3. Containment Pressure--High-High	≤ 8.0 psig	≤ 8.9 psig

TABLE 3.3-4 (Continued)

ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
4. STEAM LINE ISOLATION		
a. Manual	Not Applicable	Not Applicable
b. Automatic Actuation Logic	Not Applicable	Not Applicable
c. Containment Pressure-- Intermediate-High-High	≤ 3.0 psig	≤ 3.9 psig
d. Steamline Pressure-Low	≥ 500 psig steam line pressure	≥ 488 psig steam line pressure
e. High Negative Steam Pressure Rate	≤ 100 psi with a time constant ≥ 50 seconds	≤ 127 psi with a time constant ≥ 50 seconds
5. TURBINE TRIP AND FEEDWATER ISOLATION		
a. Steam Generator Water Level High-High	$\leq 75\%$ of narrow range instrument span each steam generator	≤ 76.9 of narrow range instrument span each steam generator
6. LOSS OF POWER		
a. 1. 4.16kv Emergency Bus Undervoltage (Loss of Voltage) (Trip Feed)	$\geq 75\%$ of nominal bus voltage with a 1 ± 0.1 second time delay	$\geq 73\%$ of nominal bus voltage with a 1 ± 0.1 second time delay
2. 4.16kv Emergency Bus (Start Diesel)	$\geq 83\% - 12$ cycles	$\geq 81\%$ of nominal bus voltage
b. 4.16kv Emergency Bus Undervoltage (Degraded Voltage)	$\geq 90\%$ of nominal bus voltage with a 90 ± 5 second time delay	$\geq 88\%$ of nominal bus voltage with a 90 ± 5 second time delay
c. 480v Emergency Bus Undervoltage (Degraded Voltage)	$\geq 90\%$ of nominal bus voltage with a 90 ± 5 second time delay	$\geq 88\%$ of nominal bus voltage with a 90 ± 5 second time delay

TABLE 3.3-4 (Continued)

ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
7. AUXILIARY FEEDWATER		
a. Steam Generator Water Level-low-low	$\geq 12\%$ of narrow range instrument span each steam generator	$\geq 10.7\%$ of narrow range instrument span each steam generator
b. Undervoltage - RCP	≥ 2750 volts RCP bus voltage	≥ 2687 volts RCP bus voltage
c. S.I.	See 1 above (all SI Setpoints)	
d. Emergency Bus Undervoltage	≤ 3350 volts	≤ 3325 volts
e. Trip of Main Feedwater Pumps	Not Applicable	Not Applicable

REACTOR TRIP SYSTEM INSTRUMENTATION TRIP SETPOINTS

FUNCTIONAL UNIT	TRIP SETPOINT	ALLOWABLE VALUES	
		Not Applicable	
1. Manual Reactor Trip	Not Applicable	Not Applicable	
2. Power Range, Neutron Flux	Low Setpoint - $\leq 25\%$ of RATED THERMAL POWER High Setpoint - $\leq 109\%$ of RATED THERMAL POWER	Low Setpoint - $\leq 27.3\%$ of RATED THERMAL POWER High Setpoint - $\leq 111.3\%$ of RATED THERMAL POWER	
3. Power Range, Neutron Flux, High Positive Rate	$\leq 5\%$ of RATED THERMAL POWER with a time constant ≥ 2 seconds	$\leq 6.3\%$ of RATED THERMAL POWER with a time constant ≥ 2 seconds	
4. Power Range, Neutron Flux, High Negative Rate	$\leq 5\%$ of RATED THERMAL POWER with a time constant ≥ 2 seconds	$\leq 6.3\%$ of RATED THERMAL POWER with a time constant ≥ 2 seconds	
5. Intermediate Range, Neutron Flux	$\leq 25\%$ of RATED THERMAL POWER	$\leq 31.1\%$ of RATED THERMAL POWER	
6. Source Range, Neutron Flux	$\leq 10^3$ counts per second	$\leq 1.4 \times 10^3$ counts per second	
7. Overtemperature AT	See Note 1	See Note 3	
8. Overpower AT	See Note 2	See Note 3	
9. Pressurizer Pressure--Low	≥ 1945 psig	≥ 1934 psig	
10. Pressurizer Pressure--High	≤ 2385 psig	≤ 2344 psig	
11. Pressurizer Water Level--High	$\leq 92\%$ of instrument span	$\leq 93.9\%$ of instrument span	
12. Loss of Flow	$\geq 90\%$ of design flow* per loop	$\geq 89.2\%$ of design flow* per loop	

*Design flow is 88,500 gpm per loop.

TABLE 2.2-1 (Continued)

REACTOR TRIP SYSTEM INSTRUMENTATION TRIP SETPOINTS

FUNCTIONAL UNIT	TRIP SETPOINT	ALLOWABLE VALUES
13. Steam Generator Water Level-Low-Low	$\geq 12\%$ of narrow range instrument span-each steam generator	≥ 10.7 of narrow range instrument span-each steam generator
14. Steam/Feedwater Flow Mismatch and Low Steam Generator Water Level	$\leq 40\%$ of full steam flow at RATED THERMAL POWER coincident with steam generator water level $\geq 25\%$ of narrow range instrument span-each steam generator	≤ 43.4 of full steam flow at RATED THERMAL POWER coincident with steam generator water level ≥ 23.1 of narrow range instrument span-each steam generator
15. Undervoltage-Reactor Coolant Pumps	≥ 2750 volts-each bus	≥ 2687 volts-each bus
16. Underfrequency-Reactor Coolant Pumps	≥ 57.5 Hz - each bus	≥ 57.4 Hz - each bus
17. Turbine Trip A. Auto stop oil pressure B. Turbine Stop Valve	45 psig $\geq 1\%$ open	± 5 psig $\geq 1\%$ open
18. Safety Injection Input from ESF	Not Applicable	Not Applicable
19. Reactor Coolant Pump Breaker Position Trip	Not Applicable	Not Applicable
20. Reactor Trip System Interlocks		
A. Intermediate Range Neutron Flux, P-6	$\geq 1 \times 10^{-10}$ Amps	$\geq 6 \times 10^{-11}$ Amps

Proposed Wording

TABLE 2.2-1 (Continued)
 REACTOR TRIP SYSTEM INSTRUMENTATION TRIP SETPOINT

FUNCTIONAL UNIT	TRIP SETPOINT	ALLOWABLE VALUES
20. (Continued)		
B. Power Range Neutron Flux, P-8	≤ 30% RATED THERMAL POWER	32.3 ≤ 31% RATED THERMAL POWER
C. Power Range Neutron Flux, P-9	≤ 49% RATED THERMAL POWER	51.3 ≤ 51% RATED THERMAL POWER
D. Power Range Neutron Flux, P-10 (Input to P-7)	10% RATED THERMAL POWER	≥ 7.7% and ≤ 12.3% → 9% and ≤ 12% RATED THERMAL POWER
E. Turbine Impulse Chamber Pressure, P-13 (Input to P-7)	≤ 10% of RTP Turbine Impulse Pressure Equivalent	≤ 12.3% of RTP Turbine Impulse ≤ 72-PSIG Pressure Equivalent

TABLE 2.2-1 (Continued)

REACTOR TRIP SYSTEM INSTRUMENTATION TRIP SETPOINTS

NOTATION

NOTE 2: $\text{Overpower } \Delta T \left(\frac{1}{1+\tau_4 S} \right) \leq \Delta T_0 \left[K_4 - K_5 \left(\frac{\tau_3 S}{1+\tau_3 S} \right) \left(\frac{1}{1+\tau_5 S} \right) T - K_6 \left[T \left(\frac{1}{1+\tau_5 S} \right) - T'' \right] - f(\Delta I) \right]$

where: ΔT_0	=	Indicated ΔT at RATED THERMAL POWER
T	=	Average temperature, °F
T''	=	Indicated T_{avg} at RATED THERMAL POWER $\leq 576.3^\circ\text{F}$
K_4	=	1.07
K_5	=	0.02/°F for increasing average temperature
K_6	=	0.00128 for $T > T''$; $K_6 = (0)$ for $T \leq T''$
$\frac{\tau_3 S}{1+\tau_3 S}$	=	The function generated by the rate lag controller for T_{avg} dynamic compensation
τ_3	=	Time constant utilized in the rate lag controller for T_{avg} , $\tau_3 = 10$ secs.
$\frac{1}{1+\tau_4 S}$	=	Lag compensator on measured ΔT
τ_4	=	Time constant utilized in the lag compensator for ΔT , $\tau_4 \leq 2$ secs.
$\frac{1}{1+\tau_5 S}$	=	Lag compensator on measured T_{avg}
τ_5	=	Time constant utilized in the lag compensator for $T_{avg} \leq 2$ secs.
S	=	Laplace transform operator.
$f(\Delta I)$	=	0 for all ΔI

NOTE 3: The channel's maximum trip point shall not exceed its computed trip point by more than 4 percent.

3.4

TABLE 3.3-4

ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

FUNCTIONAL UNIT	TRIP SETPOINT	ALLOWABLE VALUES
1. SAFETY INJECTION, TURBINE TRIP AND FEEDWATER ISOLATION		
a. Manual Initiation	Not Applicable	Not Applicable
b. Automatic Actuation Logic	Not Applicable	Not Applicable
c. Containment Pressure--High	≤ 1.5 psig	2.4 < 2.0 psig
d. Pressurizer Pressure-Low	> 1845 psig	1830 > 1835 psig
e. Steamline Pressure-Low	> 500 psig steam line pressure	488 > 400 psig steam line pressure

Proposed Wording

TABLE 3.3-4 (Continued)

ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

FUNCTIONAL UNIT	TRIP SETPOINT	ALLOWABLE VALUES
1.1 SAFETY INJECTION-TRANSFER FROM INJECTION TO THE RECIRCULATION MODE		
a. Manual Initiation	Not Applicable	Not Applicable
b. Automatic Actuation Logic Coincident with Safety Injection Signal	Not Applicable	Not Applicable
c. Refueling Water Storage Tank Level-Low	$18'8\frac{1}{2}"$ $19'2\frac{1}{2}" \pm 0'6"$	$\geq 18'2\frac{1}{2}"$ and $\leq 19'2\frac{1}{2}"$ $19'2\frac{1}{2}" \pm 1'0"$
d. Refueling Water Storage Tank Level - Auto QS Flow Reduction	$9'6"$ $11'0" \pm 3"$	$\geq 9'0"$ and $\leq 10'0"$ $11'0" \pm 6"$

Proposed Wording

TABLE 3.3-4 (Continued)
ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
2. CONTAINMENT SPRAY		
a. Manual Initiation	Not Applicable	Not Applicable
b. Automatic Actuation Logic	Not Applicable	Not Applicable
c. Containment Pressure--High-High	8.0 ≤ +10 psig	8.9 ≤ +11 psig
3. CONTAINMENT ISOLATION		
a. Phase "A" Isolation		
1. Manual	Not Applicable	Not Applicable
2. From Safety Injection Automatic Actuation Logic	Not Applicable	Not Applicable
b. Phase "B" Isolation		
1. Manual	Not Applicable	Not Applicable
2. Automatic Actuation Logic	Not Applicable	Not Applicable
3. Containment Pressure--High-High	8.0 ≤ +10 psig	8.9 ≤ +11 psig

BEAVER VALLEY - UNIT 1

3/4 3-23

Proposed Wording

TABLE 3.3-4 (Continued)

ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

FUNCTIONAL UNIT	TRIP SETPOINT	ALLOWABLE VALUES
4. STEAM LINE ISOLATION		
a. Manual	Not Applicable	Not Applicable
b. Automatic Actuation Logic	Not Applicable	Not Applicable
c. Containment Pressure-- Intermediate-High-High	3.0 ≤ 5.0 psig	3.9 ≤ 5.5 psig
d. Steamline Pressure-Low	≥ 500 psig steam line pressure	488 ≥ 400 psig steam line pressure
e. High Negative Steam Pressure Rate	≤ 100 psi with a time constant ≥ 50 seconds	127 ≤ 110 psi with a time constant ≥ 50 seconds
5. TURBINE TRIP AND FEEDWATER ISOLATION		
a. Steam Generator Water Level High-High	≤ 75% of narrow range instrument span each steam generator	76.9 ≤ 76% of narrow range instrument span each steam generator
6. LOSS OF POWER		
a. 1. 4.16kv Emergency Bus Undervoltage (Loss of Voltage) (Trip Feed)	≥ 75% of nominal bus voltage with a 1 ± 0.1 second time delay	73 ≥ 74% of nominal bus voltage with a 1 ± 0.1 second time delay
2. 4.16kv Emergency Bus (Start Diesel)	≥ 83% - 12 cycles	≥ 81% of nominal bus voltage
b. 4.16kv Emergency Bus Undervoltage (Degraded Voltage)	≥ 90% + 3% 90% - 0% of nominal bus voltage with a 90 ± 5 second time delay	88 ≥ 89% of nominal bus voltage with a 90 ± 5 second time delay
c. 480v Emergency Bus Undervoltage (Degraded Voltage)	≥ 90% + 3% 90% - 0% of nominal bus voltage with a 90 ± 5 second time delay	88 ≥ 89% of nominal bus voltage with a 90 ± 5 second time delay

Proposed Wording

Proposed Wording

TABLE 3.3-4 (Continued)

ENGINEERED SAFETY FEATURE ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

FUNCTIONAL UNIT	TRIP SETPOINT	ALLOWABLE VALUES
7. AUXILIARY FEEDWATER		
a. Steam Generator Water Level-low-low	$\geq 12\%$ of narrow range instrument span each steam generator ≥ 2750 volts RCP bus voltage	≥ 10.7 of narrow range instrument span each steam generator ≥ 2687 volts RCP bus voltage
b. Undervoltage - RCP		
c. S.I.	See 1 above (all SI Setpoints)	
d. Emergency Bus Undervoltage	≤ 3350 volts	≤ 3325 volts
e. Trip of Main Feedwater Pumps	Not Applicable	Not Applicable

Proposed Unit 1 Technical Specification Change No. 1A-156
Safety Analysis

1920 psig was the most limiting value from the LOCA (1845 psig) and non-LOCA (1920 psig) for the setpoint development. The most limiting value is determined by the value closest to the operating pressure.

- The high pressurizer pressure reactor trip is actively credited in the Loss of Load Transient. A safety analysis setpoint value of 2410 psig was assumed for this event. A safety analysis setpoint value of 2436 psig is required in order to maintain the current technical specification setpoint of 2385 psig. Based on a performed safety evaluation, an increase from 2410 psig to 2436 psig will have a minimal impact on the FSAR analysis results.

- As part of the safety analyses, the 80% value was used in the Feedwater System Malfunction transient event.

- A specific setpoint value is not assumed in the Loss of Flow analysis. The analysis assumes RCP bus voltage is lost at time zero and RCP begins to coastdown. Reactor trip is assumed to occur on RCP undervoltage following an appropriate delay time. A value greater than 60% of RCP bus voltage is assumed for setpoint development. This is based on maintaining a minimum voltage for RCP motor torque and full loop flow.

The following is a discussion on changes to Table 2.2-1 and Table 3.3-4 of the Technical Specifications:

- The Containment Pressure High-High and Intermediate High-High safety analysis limits are 10 psig and 5.0 psig respectively. The current trip settings for these items have not addressed instrument error. The proposed change revises these trip setpoints and allowable values to ensure that when instrument channel inaccuracies are considered the safety analysis limits will not be exceeded. The current Containment Pressure-High trip setting has also not considered instrument error. However, a safety analysis has been performed which revises the safety analysis limit for Containment Pressure-High from 1.5 psig to 3.5 psig. With this revised safety analysis limit the current trip setpoint of 1.5 psig is adequate to ensure that with instrument inaccuracies considered the safety analysis limit will not be exceeded.

2. Turbine Impulse Chamber Pressure P-13 Interlock

The proposed change revises the Turbine Impulse Chamber Pressure P-13 Interlock. This change does not change the actual trip setting but simply revises the method of presenting the required setting from impulse pressure in psig to % rated thermal power equivalent.

3. 4KV and 480V Emergency Buses Undervoltage

The proposed change also revises the 4KV and 480V Emergency Buses Undervoltage setting for degraded voltage. The revised setpoints are made to be consistent with the methodology used to calculate instrument channel inaccuracies and does not impact on the margin to any safety analyses.

4. RWST Level Low and QS Flow Reduction

The RWST Level Low and QS Flow Reduction setpoints were revised in WCAP 11419 from $19'2\frac{1}{2}'' \pm 6''$ and $11'0'' \pm 3''$ to $\geq 19'2\frac{1}{2}''$ and $\geq 11'0''$ respectively. These setpoints would ensure that the accident analysis requirements for containment depressurization are met. However, as discussed in the BVPS Unit 1 FSAR (Section 6.3.2.1), the RWST Level-Low setpoint also ensures that the containment sump level will be sufficient to provide adequate NPSH for the Low Head Safety Injection (LHSI) Pumps after the transfer to recirculation. Therefore, this setpoint must also ensure that a minimum amount of water is directed into the containment and sump prior to transfer to recirculation. This setpoint, therefore, has both an upper and lower safety limit. In addition, the RWST Level channel inaccuracies (channel statistical allowance of 2.6% or 17") calculated in WCAP 11419 were greater than the inaccuracies assumed in our current accident analyses ($\pm 1'$ for transfer to recirculation and $\pm 6''$ for QS flow reduction). The affected accident analyses for containment depressurization and LHSI Pumps NPSH requirements were therefore, re-analyzed using revised safety limits of $\pm 18''$ for both setpoints. The results of this analysis indicated that the containment depressurization design basis and the LHSI Pumps NPSH requirements continues to be met. However, to satisfy the analysis requirements it was necessary to lower the RWST Level-Low setpoint by 6" to $18'8\frac{1}{2}''$ and the RWST Level Auto QS Flow Reduction setpoint by $1\frac{1}{2}'$ to $9'6''$. With this re-analysis, these setpoints are now within the bounds of our safety limits when the instrument channel inaccuracies calculated by WCAP 11419 are considered. New allowable values were also calculated for these setpoints using the methodology provided in WCAP 11419. The Unit 1 FSAR will be updated to include the revised analysis results.

5. Tables 2.2-1 and 3.3-4 Allowable Values

Various Allowable Values of Tables 2.2-1 and 3.3-4 are revised in the proposed change. These changes are based on the results of WCAP 11419 and the methods described in that report for calculating Allowable Values. Some are more restrictive and some less restrictive than what are currently in the Technical Specifications. The Allowable Values provided will ensure that if the measured setpoint value is less than the Allowable Value the instrument channel is well within the accuracy allowance.

In summary, the proposed setpoint and Allowable Value changes will ensure that the trip setpoints assumed in the Beaver Valley Unit 1 safety analyses will be valid when the required channel instrument inaccuracies are considered. The method for determining these inaccuracies is consistent with previously approved methods for calculating instrument error. These proposed changes are therefore considered safe.

A T T A C H M E N T C

No Significant Hazard Evaluation

Proposed Technical Specification Change No. 1A-156

Basis for Proposed No Significant Hazards Consideration Determination: The Commission has provided standards for determining whether a significant hazards consideration exists (10 CFR 50.92(c)). A proposed amendment to an operating license for a facility involves no significant hazards consideration if operation of the facility in accordance with the proposed amendment would not (1) involve a significant increase in the probability or consequences of an accident previously evaluated; (2) create the possibility of a new or different kind of accident from any accident previously evaluated; or (3) involve a significant reduction in a margin of safety.

The proposed change does not involve a significant hazards consideration because:

- 1) The basis of the Reactor Trip System and ESF Actuation System trip settings are the values assumed in the Safety Analyses. The proposed changes to the trip settings and Allowable Values ensures that these safety analysis limits are maintained when the appropriate allowances for instrument channel inaccuracies are considered. Therefore, the proposed changes do not involve a significant increase in the probability or consequences of an accident previously evaluated.
- 2) The proposed changes are made to ensure that the safety analyses limits assumed in the analysis remain valid by assigning the appropriate values for instrument channel inaccuracies to the trip settings and allowable values. This change does not involve any plant equipment or design change. Therefore, the probability of an accident or a malfunction of a different type than previously evaluated would not be created.
- 3) This change will not affect the assumptions or consequences of any safety analysis presented in the FSAR. Therefore, the change will not involve a significant reduction in a margin of safety.

Based on the above considerations, it is proposed to characterize the change as involving no significant hazards consideration.